

PORTFOLIO PROJECT · AI & AUTOMATION

Construction Project Intake & Crew Dispatcher

A multi-branch intelligent automation system that takes a construction project from a Tally form submission all the way to a fully assigned, deadline-set, client-delivered job — and keeps managing it for its entire lifecycle, including staff turnover, automatic reassignment, and finished-output delivery. Almost the whole pipeline runs without a human in the loop.

WHAT IT DOES

Automates project intake, AI-driven document analysis, recursive estimator matching, Monday.com synchronisation, deadline calculation, Slack notification, and full lifecycle management — including automatic reassignment when staff resign.

TOOLS USED

n8n · Tally · Webhooks & APIs · AI Agent (Google Gemini) · Google Drive · Google Sheets · Monday.com · Slack · JavaScript

WHO BENEFITS

Construction managers, estimation teams, and operations leads looking to eliminate manual admin and stop projects from being dropped during staff changes.

KEY FEATURES

Automatic folder creation · Recursive estimator availability loop · Self-healing roster sync · Resignation-triggered project recovery · Webhook-triggered completion handoff · Cross-platform sync

BUILT BY

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What Is This Project?

In construction estimation, every new project starts the same way: a client submits plans and requirements, and the manual work begins — organising files, reading documents, finding an estimator who is genuinely free, assigning the work, setting a deadline, and telling the team. And that's before accounting for what happens when a staff member resigns mid-project.

This system replaces almost that entire chain. From the moment a **Tally** form fires a **webhook**, a multi-branch **n8n** workflow organises the documents, calls an **AI Agent** built on Google Gemini to extract the data, continuously monitors staff availability until someone is free, assigns the project in **Monday.com** over its API, calculates the due date, syncs Google Sheets, and briefs the team on Slack — then keeps watching the project for its entire lifecycle, automatically recovering any work left behind when an employee resigns.

4

Pipeline Phases

Intake, Allocation, Handoff & Lifecycle Management

3

Manual Touchpoints

Add staff to Monday & Slack, mark a resignation, upload final output

Auto

Recovery Loop

Re-assigns work automatically when staff resign

Live

Status Tracking

Monday.com, Sheets & Slack updated in real time

Why a Civil Engineer built this: Having worked across project engineering and estimation coordination, the pain of manual intake, dispatching, and chasing staff availability is deeply familiar. This system was designed around how a real construction estimation team actually operates — including the messy parts most automations ignore, like someone leaving mid-project.

The 4 Phases — System Overview

The system runs as four connected phases, each handing off cleanly to the next:

PHASE 1

Project Intake & Infrastructure

Captures the Tally form submission via webhook, builds the Drive folder structure, sorts documents by type, and calls an AI Agent (Google Gemini) to extract structured cost and specification data from the uploaded files.

PHASE 2

State-Machine & Resource Allocation

Checks live staff availability against Monday.com and Google Sheets via API, assigns immediately if someone is free, or holds the project in a Pending queue and keeps re-checking every minute until a match is confirmed.

PHASE 3

Synchronisation & Team Handoff

Assigns the project in Monday.com, calculates the due date in JavaScript, syncs Google Sheets, and sends a full project briefing to the right person on Slack.

PHASE 4

Lifecycle Management & Roster Integrity

Continuously monitors the employee roster against the live board. The moment someone is deactivated on Monday.com and Slack, any active project they were holding is automatically recovered, reverted to Unassigned, and re-queued — with its original cost and location data intact.

From Form Submission to Structured, AI-Ready Data

Everything begins with a single Tally form submission. From that moment, a webhook hands the payload to n8n, which takes full control of organising, categorising, and preparing every project document — so the estimation team receives clean, structured numbers instead of a pile of raw PDFs.

1

A project form is submitted & a webhook fires

The workflow triggers the instant a client submits the Tally intake form — capturing the project name, client name, contact details, location, and any uploaded plan or requirement files, delivered straight into n8n over a webhook.

2

A dedicated Google Drive folder is created automatically

The system creates a structured folder for the project via the Drive API, with separate sub-folders for plans and technical requirements — every document has a home from the very first second.

3

Documents are sorted and uploaded into the right folders

Plan files and requirement files are routed automatically into their correct sub-folder — no manual sorting, no misplaced attachments.

4

An AI Agent reads and extracts data from the uploaded files

An AI Agent built on Google Gemini is given the uploaded documents and instructed to extract project type, construction classification, location, floor area, and an estimated unit-cost range — the numbers an estimator would otherwise spend hours pulling out by hand.

5

The Master List in Google Sheets is updated

Project, client, and document details are logged to a central spreadsheet via the Sheets API — the single source of truth every later phase, including resignation recovery, reads from.

Why this matters: By the time Phase 1 finishes, every document has already been read and converted into structured numbers. The estimator skips straight to decision-making instead of spending the first hour figuring out what the project even needs.

A Loop That Never Lets a Project Sit Idle

Once the project data is structured, the system needs to find the right person for it. People aren't always free — they're mid-project, on leave, or already at capacity. This phase handles that with a persistent matching loop instead of a single point-in-time check.

6

The project is created on the Monday.com board

A new item is created via the Monday.com API the moment intake completes, with its project reference link and due-date placeholder attached.

7

Staff availability is checked against the live employee roster

Google Sheets is queried in real time for who is marked Available versus Assigned — never a cached snapshot.

8

If someone is free, they're matched and assigned immediately

The first available, qualified employee is locked in, and their details are passed straight to the handoff phase.

9

If nobody is available, the project waits — and the system keeps trying

Instead of failing or sitting forgotten, the project drops into a Pending queue with its AI cost and location data attached. A schedule trigger re-checks availability every minute until someone is confirmed free.

10

The moment someone is free, the queue is cleared

As soon as an employee's status flips to Available — whether from finishing a job or a resignation freeing up a slot — the oldest Pending project is assigned without anyone having to trigger it manually.

Why a recurring loop instead of a single check: A one-time availability check fails the moment everyone happens to be busy. The recurring loop means the instant someone becomes free, the waiting project is picked up — with zero manual follow-up.

Assignment, Deadlines, and Instant Team Notification

With the right estimator identified, this phase connects every platform over its API, makes the formal assignment, calculates when the work is due, and puts everything in front of the team in the tools they already use daily.

11

The project is assigned on Monday.com

A direct API call sets the owner, status, and reference link on the board item — giving the whole team instant, live visibility.

12

The due date is calculated in JavaScript

A Code node calculates an accurate turnaround date the moment of assignment, then writes it straight back to Monday.com via API.

13

Google Sheets is updated with the final status

Assigned employee, due date, AI-extracted specs, and cost range are all written back to the Master List — one place to check on any project, instantly.

14

The Slack member directory is queried for the right recipient

The assigned employee's Monday.com profile is matched against a self-maintained Slack member cache to find their exact Slack ID — built to minimise repeated Slack API calls and stay well inside rate limits.

15

A full project briefing is posted and DM'd

The team channel and the assigned employee both receive the project name, location, AI cost assessment, and due date — clearly flagged as an AI estimate that still needs human verification.

End result so far: From a single form submission, the system has created a folder, read every document with AI, found the right estimator, assigned the project, set the deadline, and briefed the team — without a single manual step.

What Happens After Assignment — And When Staff Leave

Most intake automations stop the moment a task is assigned. This system keeps watching. A roster sync runs every minute in the background, comparing the live Monday.com board against the employee database — so the system always knows who is actually still on the team, and what they're holding.

16

A new hire is added once — the rest syncs itself

The only manual step is inviting the new employee to the Monday.com board and Slack. From there, the roster sync detects them automatically and even self-heals stale records — for example, correcting a placeholder email-as-name once they finish setting up their profile.

17

A resignation is detected the moment it happens

The only manual step here is deactivating or removing the employee on Monday.com and Slack. The system picks it up on the very next one-minute cycle — no one has to remember to update a spreadsheet.

18

Their active project is found and recovered — automatically

Any project still owned by the resigned employee (excluding work already Finished) is located, reverted to Unassigned on the board, and dropped straight back into the Pending queue, carrying its original location and AI cost data with it.

19

The team is notified, and the queue takes over

A Slack message confirms the project has been freed up and re-queued; Phase 2's availability loop then re-assigns it the moment another estimator is free — with no human intervention at any point.

20

Finished work is routed straight to the client

The only manual step left is uploading the deliverable and marking the project Finished on Monday.com. That status change fires a webhook: the system locates the correct Drive folder by project name, creates a "Finished Output" sub-folder, uploads the file, and emails the client a styled notification with a direct link to review it.

Why this phase exists: Staff turnover is the part every other intake automation ignores. Without it, a resignation would leave a project frozen on a dead account forever. With it, the workflow recovers the work in under a minute and gets it back in front of an available estimator — with zero data loss and zero manual cleanup.

The Tools Working Together

Nine tools are connected across this system. Each plays a specific, non-overlapping role:

n

n8n — The automation engine

Powers all four phases — connecting every tool, running the logic, managing the recurring availability and roster-sync loops, and orchestrating the whole pipeline end to end.

T

Tally — The intake form

Captures the project, client, and document details that kick off the entire pipeline, and fires the webhook that triggers Phase 1.

W

Webhooks & APIs — The integration layer

Real-time triggers (form submission, Monday.com status changes) and direct REST calls are what let every platform in this system talk to each other instantly, with no polling.

G

AI Agent (Google Gemini) — The intelligent document reader

Reads uploaded construction plans and specs and extracts project type, classification, location, floor area, and cost estimates in seconds.

G

Google Drive — The document organiser

Every project gets its own folder structure, automatically separated into plans and requirements, with a "Finished Output" sub-folder created on completion.

G

Google Sheets — The master registry

Tracks every project, every employee's live status, and the Pending queue — always in sync with Monday.com and updated automatically after every change.

M

Monday.com — The project board

Tasks are created, assigned, reverted, and tracked via direct API calls — the board reflects every change in real time, including when work is freed up by a resignation.

J

JavaScript — The logic engine

Custom Code nodes calculate due dates, diff the live roster against the employee database to detect new hires and resignations, and filter board items to find orphaned projects.

S

Slack — The team notification channel

Posts new-project, pending, reassignment, and resignation alerts to the team channel, and DMs the assigned estimator directly — backed by a self-maintained member cache so names always resolve to the right person.

Who Benefits from This System?

Built with three groups in mind — each gets something different out of it:

Construction Managers

Real-time project tracking without checking in with anyone. Every project is visible on Monday.com the moment it's assigned, with a due date already set — and it stays visible even if the assigned person leaves the team.

Estimation Teams

Immediate access to AI-extracted cost and spec data the moment a project arrives. No more spending hours reading documents manually before the real estimation work can even begin.

Operations

Drastically reduced administrative overhead — no manual reassignment when someone resigns, no chasing availability, no copy-pasting between platforms beyond the three light touchpoints the system still needs.

Why This Matters



No project ever gets lost in a pile of emails

Every submission is instantly captured, organised, and tracked in a live system.



AI does the reading so your team doesn't have to

An AI Agent built on Google Gemini extracts cost and spec data in seconds — work that would otherwise take hours.



The right estimator is always found, no matter how busy the team is

The recurring availability loop means no project ever stalls.



No project is ever orphaned by staff turnover

A resignation automatically frees and re-queues whatever that person was holding — the feature most automations leave out entirely.



Every platform stays in sync automatically

Monday.com, Google Sheets, Google Drive, and Slack all reflect the same accurate information in real time.



The team gets everything they need the moment a project is assigned

The Slack notification includes the AI cost estimate, location, and due date — ready to act on immediately.